

ASE 420 Tools - Obsidian

Obsidian

- Obsidian is a personal knowledge management app — sometimes called **PKM** — built on plain Markdown files.
- Notes remain ordinary `.md` files that you can organize and search.
- **In ASE 420, use it to connect course concepts with your project's design decisions** — for example, linking a pattern you learned to the reason your team chose it.

| Install Obsidian

- Download it from obsidian.md.
- Install the desktop application for your operating system.
- You do not need an account to keep a local vault.

Vault Basics

- A **vault** is a folder containing notes and attachments.
- Create one vault for ASE 420 in a location you can find and back up.
- Use a small folder structure rather than many nested folders.

A Simple ASE 420 Vault

ASE420/
Course Notes/
Project/
Attachments/

- Start with this basic structure and get comfortable using Obsidian day to day — you can expand it later as your workflow develops.

Markdown Notes

- Obsidian notes are stored as plain text `.md` files for long-term portability.
- Format structured content using headings, lists, code blocks, and math equations.
- Link concepts using double brackets (`[[Link]]`) to connect related notes.
- Keep one main idea or meeting in each note, with a descriptive file name.

Strategy Pattern

- Problem: behavior changes at runtime
- Project example: scoring strategy

Questions

- When should we prefer composition?

A Simple Note Workflow

1. Capture an idea, question, or decision quickly.
2. Add a clear title and short explanation.
3. Link it to related concepts or project notes.
4. Review and refine it after class or a team meeting.

Link Your Notes

- Create an internal link with `[[Note Name]]`.
- Link a section with `[[Note Name#Section]]`.
- Use backlinks to see which notes refer to the current note.

The implementation uses the `[[Strategy Pattern]]`.
See `[[Team Meeting 03#Decisions]]`.

| Why Linking Helps

- A folder tells you where a note is stored.
- A link tells you how two ideas are related.
- Connect requirements, design patterns, code decisions, and test results.
- Add links when they help you navigate; do not link every word.

Attachments and Files

- First, set Obsidian's attachment folder to `Attachments/` (Settings → Files and Links) so dropped files land in one place.
- Use `![[file]]` to **embed** a file inline in the note; use `[[file]]` to **link** to it without embedding.

- Drag an image or PDF into a note to embed it automatically.
- Use meaningful file names such as `tetris-class-diagram.png`.
- Keep source code in the Git repository, then link to it from notes.

```
![[tetris-class-diagram.png]]  
![[Project Requirements.pdf]]
```

Use Obsidian for ASE 420

- Create one note for each lecture topic or design pattern.
- Record project requirements and unanswered questions.
- Keep dated team-meeting notes with decisions and action items.
- Document why the team selected a design, not only what it built.
- Link project decisions back to relevant course concepts.

Suggested Project Notes

Project Home
Requirements
Architecture
Design Decisions
Team Meeting YYYY-MM-DD
Testing Notes
Retrospective

- Use **Project Home** as an index linking to the important notes.

Keeping Useful LLM Records

- Summarize verified LLM insights into linked Markdown notes rather than saving raw chat logs.
- Include: prompt, verified solution, evidence/docs, and open questions.
- Link notes to course concepts (e.g., `[[Strategy Pattern]]`).

Advanced Features

Properties

- Add structured YAML metadata at the top of a note (frontmatter).
- Use key-value pairs (e.g., `tags`, `date`, `status`, `author`) to categorize and query notes.
- Obsidian renders properties as an editable interface for easy metadata management.
- Useful once your vault has grown; not needed for your first notes.

Plugins & Automation

- Like VS Code extensions, Obsidian supports plugins that add features such as Dataview (querying notes) or Kanban boards.
- As a software engineer, you can go further and script your vault, for example writing Dataview queries or a full plugin in TypeScript/JavaScript.
- Add plugins gradually, only when you have a concrete need — not before your basic note workflow feels natural.

Sharing Notes With Your Team

- Keep the authoritative, executable **source code** in your Git repository — a note can still include a short code example for reference.
- **Shared project documentation** (architecture notes, decisions) can live in a team Obsidian vault synced through Git and GitHub, but this adds setup and merge complexity — treat it as optional.
- **Personal notes** (your own learning, LLM records) usually stay in your own local vault.

Common Cautions

- **Obsidian manages knowledge; Git manages source-code versions** — keep each tool to its role.
- Do not put secrets, passwords, or private team data in the vault.
- Avoid depending on many plugins; teammates may not have them.
- Keep attachments organized so the vault stays portable.
- Back up important notes and agree before

Build a Useful Habit

- Take short notes during class and project work.
- Afterward, summarize the key idea in your own words.
- Add links to related notes and record the next action.
- A small, reviewed set of notes is more useful than a large, disorganized vault.